Tyson Grover

E.I.T.

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KEY SKILLS

- **CAD**: SolidWorks, OnShape, Fusion360
- **FEA**: SolidWorks, Ansys, Converge CFD
- **Programming Languages**: Python, MATLAB, C++, LabVIEW, VBA
- **Manufacturing**: CNC Mill, Manual Mill, Manual Lathe, 3D Printing
- **Tools**: Excel, Word, PowerPoint

EDUCATION

Bachelor's of Science, Mechanical Engineering Aug '20 - May '26

Utah Tech University Saint George, UT

- Notable classes: Autonomous Vehicles, Modern Control Theory, Thermal System Design, Computational Fluid Dynamics, Advanced Engineering Mathematics
- GPA: 3.90

PROFESSIONAL EXPERIENCE

R&D Engineering Intern Apr '22 - Present

RAM Aviation, Space, & Defense Saint George, UT

- Developed precision control systems for hydraulic and pneumatic fluids at specified pressures and temperatures, successfully enhancing reliability and meeting all client requirements
- Enhanced temperature control accuracy and minimized solenoid valve errors by implementing a thermocouple-based PID controller with Arduino
- Improved solenoid valve cycling to sub-millisecond response using a single-board computer and LabVIEW programming, streamlining the testing process by 90%
- Engineered precision components with DFM/DFA principles and GD&T in SolidWorks, significantly increasing the efficiency of pressure and helium leakage testing tools
- Enhanced compliance and streamlined testing processes by meticulously analyzing ASTM, SAE, MIL, ASME, and ISO standards
- Assembled and tested 150+ valves for Griffin Lunar Lander, Sierra Space Dreamchaser, & Blue Origin New Glenn

Manufacturing Teaching Assistant Aug '21 - Apr '22

Utah Tech University, Department of Engineering Saint George, UT

- Led & assisted in teaching a group of 20 students in basic manufacturing principles
- Provided training on manufacturing techniques such as milling, turning, and welding for introductory engineering projects, significantly enhancing student engagement through hands-on projects like a foldable knife

PROJECTS

Drop Shock Testing Tower Aug '25 - May '26

- Creating pneumatic, electrical, braking and acceleration systems with Python and Raspberry Pi, contributing to a $20K reduction in yearly expenses
- Used equations of motion & FEA to design tower to deliver up to 600 G's for 5 ms to a solenoid valve test unit
- Manufacturing to be done using waterjet, CNC mill, manual mill & lathe, laser welding

Boat Propulsion System Jan '23 - May '23

- Constructed an innovative propulsion system for a paddleboat over a span of 4 months, significantly increasing public usability and enjoyment
- Engineered gear box and water-sealing enclosures with OnShape, enhancing design functionality through DFA/DFM principles

Volunteer

Missionary Oct '23 - Jul '25

The Church of Jesus Christ of Latter-day Saints Tempe, AZ

- Built deeper relationships and boosted individual confidence within faith-based groups through active listening, public speaking, and empathy
- Facilitated the successful execution of missionary initiatives using goal & plan setting, applying effective communication and leadership tactics to elevate performance metrics by 15%